MarsLab Multi-Instrument

Scientific Analysis Report

Region: Arcadia_Planitia_North

Center: 46.553 N, -166.257 E

Generated: 2026-02-12 16:43 UTC

Instruments Included:

SHARAD: r_3898101_001_ss19_700_a

CRISM (CNN): frt0000c702_07

MOLA DEM: Global 200m blend

SHARAD Subsurface Analysis

Detection Confidence: HIGH

Detection Rate: 100.0%

Coherent Segments: 1

Depth Estimates:

Material	er	Mean (m)	Min (m)	Max (m)
Vacuum	1.0	208.7	84.3	1675.1
Porous Ice	2.8	124.7	50.4	1001.1
Pure Water Ice	3.1	118.6	47.9	951.4
Basalt (low)	4.0	104.4	42.2	837.5
Basalt (mid)	5.0	93.3	37.7	749.1
Basalt (high)	6.0	85.2	34.4	683.9

CRISM Mineral Classification

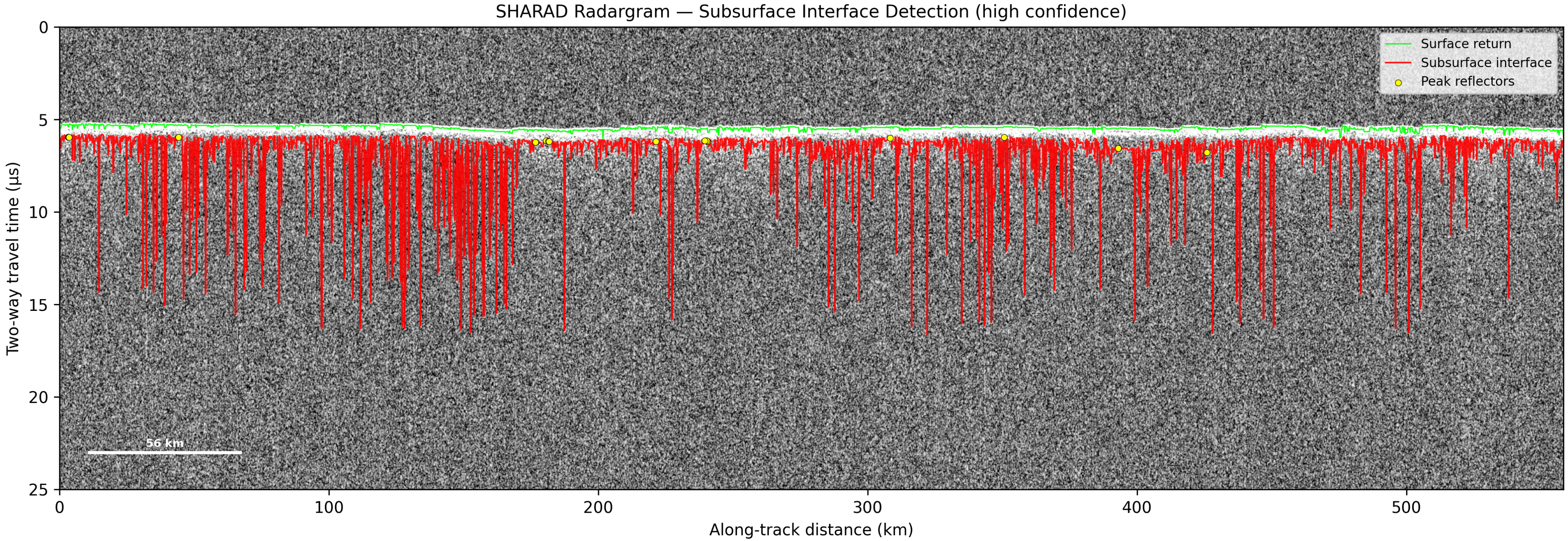
Observation: frt0000c702_07

Classified pixels: 1322 / 326400 (0.4%)

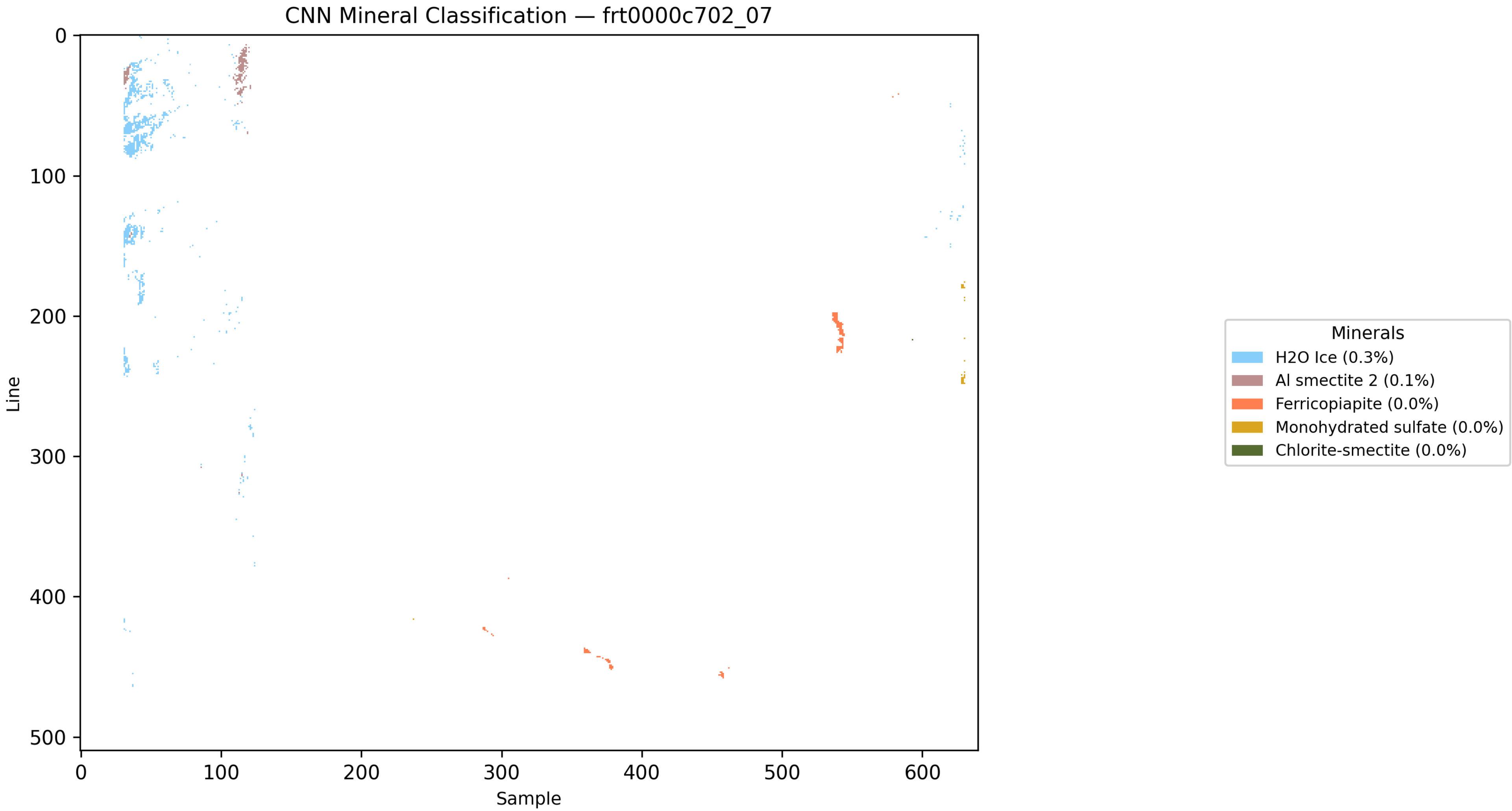
Detected Minerals:

- H2O Ice: 951 px (0.29%)
- Al smectite 2: 188 px (0.06%)
- Ferricopiapite: 153 px (0.05%)
- Monohydrated sulfate: 29 px (0.01%)
- Chlorite-smectite: 1 px (0.00%)

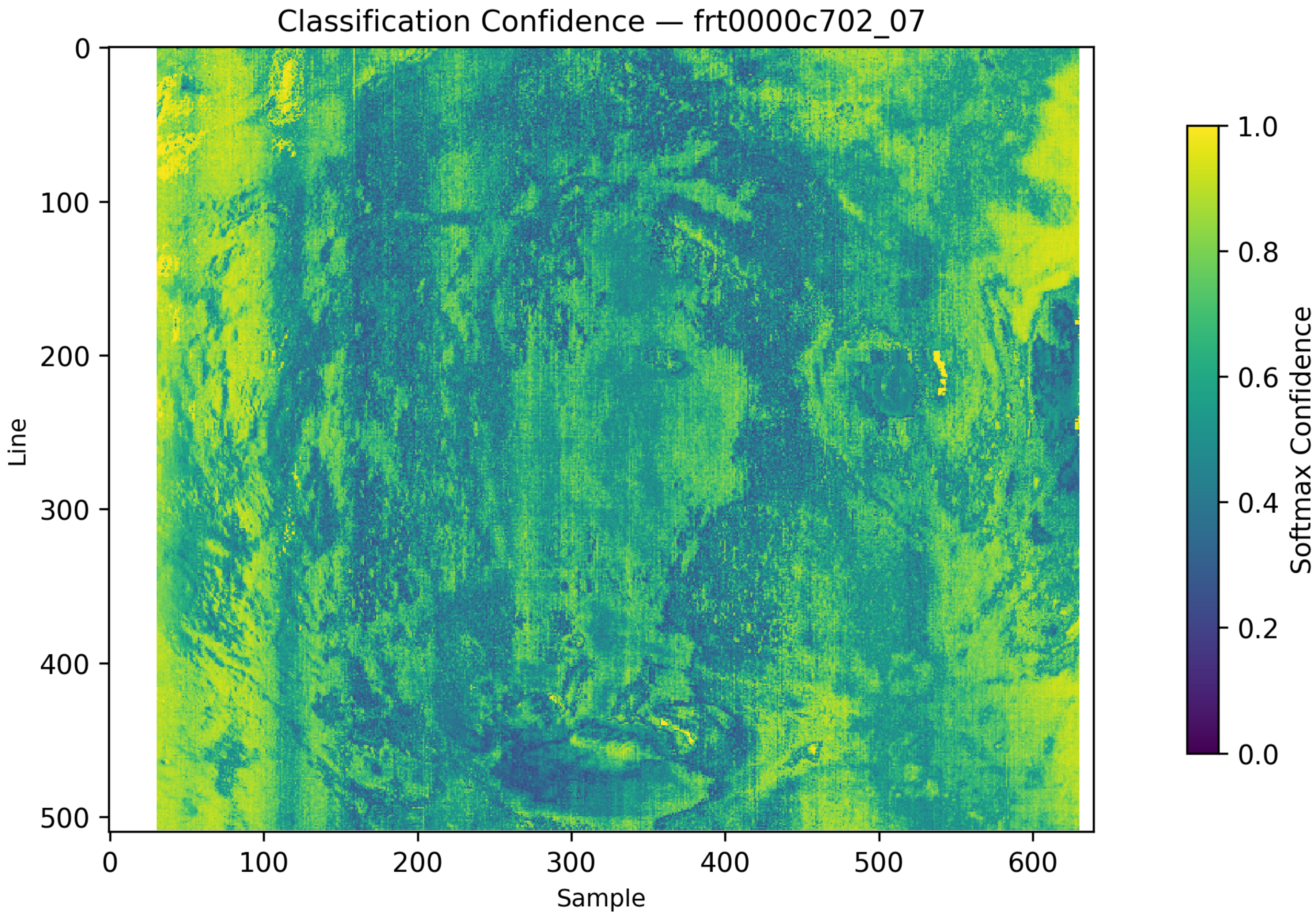
Annotated SHARAD Radargram



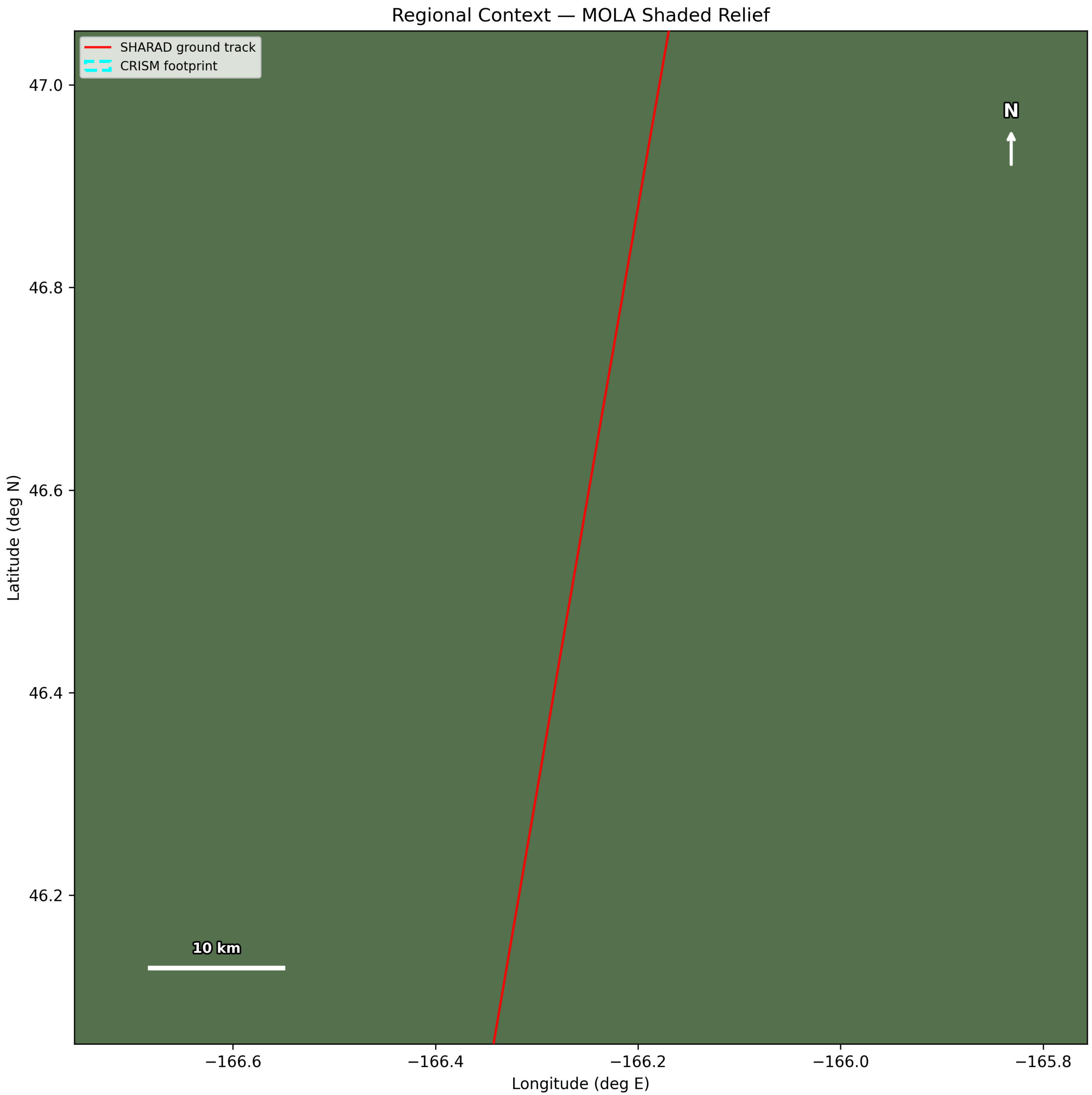
CNN Mineral Classification Map



Classification Confidence Map



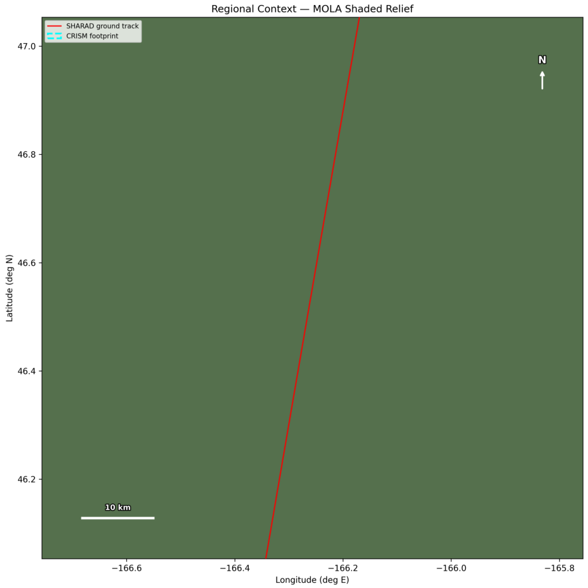
Regional Context Map (MOLA Shaded Relief)



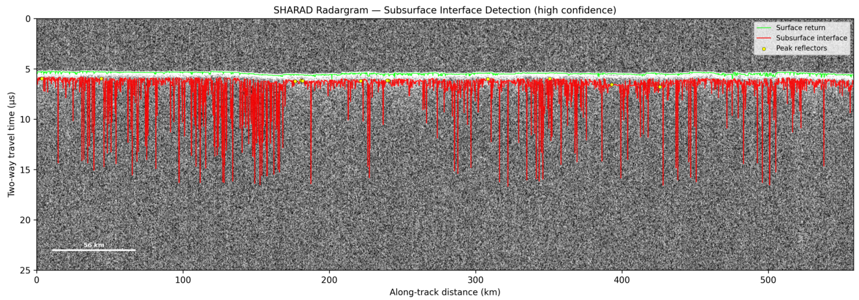
Multi-Instrument Synthesis

Multi-Instrument Synthesis — Arcadia_Planitia_North

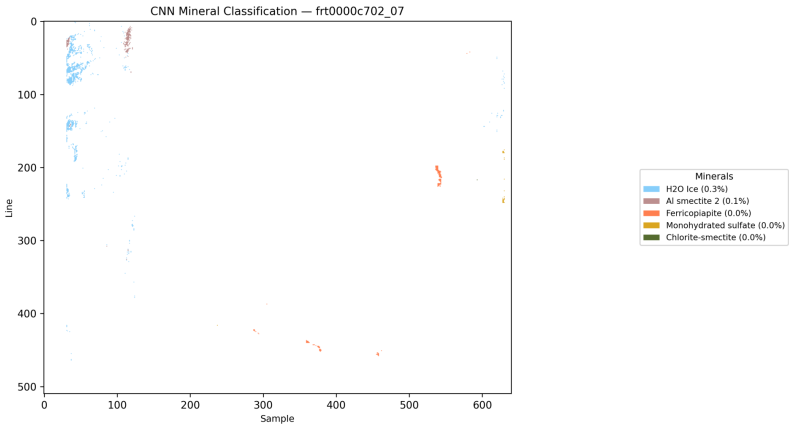
Regional Context



SHARAD Radargram



CRISM Mineral Classification



Scientific Interpretation

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SCIENTIFIC INTERPRETATION

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SUBSURFACE INTERFACE: A laterally coherent subsurface reflector was detected with HIGH confidence.

Detection rate: 100.0% of traces.

Depth (assuming water ice, er=3.1): 118.6 m mean (range: 47.9-951.4 m)

Depth (assuming basalt, er=4.0): 104.4 m mean

Depth range is physically consistent with:

- Buried ice table (if er ~ 3.1)
- Lava flow boundary (if er ~ 4-6)
- Sedimentary layer interface

MINERALOGY:

H2O Ice: 0.29%

Al smectite 2: 0.06%

Ferricopiapite: 0.05%

Monohydrated sulfate: 0.01%

Chlorite-smectite: 0.00%

=> Water/CO2 ice detected in spectral data.

Mineralogy is CONSISTENT with ice-bearing subsurface.

=> Sulfate minerals detected, indicating past aqueous activity.

=> Hydrated minerals present, suggesting water-rock interaction.

TERRAIN ASSESSMENT:

Mean slope: 0.0 deg -- FAVORABLE for landing/operations

CROSS-INSTRUMENT SYNTHESIS:

STRONG EVIDENCE: Both radar and spectral data indicate subsurface ice presence.

Assessment: Likely WATER ICE deposit.

OVERALL CONFIDENCE:

Instruments used: SHARAD, CRISM/CNN, Terrain

Multi-instrument cross-validation: HIGH confidence

Report generated: 2026-02-12 16:43 UTC

MarsLab Autonomous Scientific Report Generator